Python Assignment
Topic-1: Introduction to Python

Q1. Introduction to Python and its Features.

Ans.

Python is a very popular programming language. It is simple and easy to understand. Python is a high-level language, so we can write code in easy English format. It is also an interpreted language which means code runs line by line. Python is used in web development, data science, artificial intelligence, and automation. It supports object-oriented programming and has many built-in libraries. Python is platform independent, so it works on different systems. Because of its simplicity, beginners prefer Python.

Example:

```
print("Hello Shivani")
```

Q2. History and evolution of Python.

Ans.

Python was created by Guido van Rossum in the late 1980s. It was officially released in 1991. The main aim was to create a simple and readable language. Over time, Python became very popular because of its easy syntax. Many versions of Python were released with new features. Python 2 was used earlier, but now Python 3 is more popular. Python continues to grow because of its strong community support. Today it is one of the most widely used languages.

Q3. Advantages of using Python over other programming languages.

Ans.

Python has many advantages over other languages. It is easy to learn and use. Python requires less code compared to languages like C or Java. It is an interpreted language, so errors are easy to find. Python is platform independent and works on different systems. It provides many built-in libraries for different tasks. Python is used in many fields like web development, AI, and data science. These advantages make Python a powerful language.

Q4. Installing Python and setting up the development environment.

Ans.

To use Python, we first need to install it on our computer. We can download it from the official website. After downloading, we run the setup file and install it. It is important to select 'Add Python to PATH'. After installation, we can use tools like IDLE, VS Code, or PyCharm. These tools help in writing and running programs. We can check installation using python --version command. This confirms that Python is installed successfully.

Q5. Writing and executing your first Python program.

Ans.

Writing a Python program is very easy. We can open any editor and write code using the print() function. After saving the file with .py extension, we can run it. Python executes the code line by line and shows output. This helps beginners understand programming. The first program usually prints a message on the screen. It is a basic step to start learning Python.

Example:

```
print("This is my first Python program - Shivani")
```